SC2000/CZ2100
Probability & Statistics

Week 3

Ch 4. Bivariate Data

- Introduction to Bivariate Data
- Pearson Correlation and Covariance
- Properties of Person Correlation
- Variance Sum Law II

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- Linear combination of 2 <u>independent</u> variables X and Y

Variance of $X \pm Y$: $\sigma_{x\pm y}^2 = \sigma_x^2 + \sigma_y^2$

- If the variables X and Y are <u>correlated</u>

Variance of $X \pm Y$: $\sigma_{x\pm y}^2 = \sigma_x^2 + \sigma_y^2 \pm 2\rho\sigma_x\sigma_y$

- For computation based on a sample

$$s_{x\pm y}^2 = s_x^2 + s_y^2 \pm 2rs_xs_y$$

Eg: Students took 2 parts of a test, each worth 50 points. Part A has a variance of 25, and Part B has a variance of 49. The correlation between the test scores is 0.6.

(i) If the teacher adds the grades of the two parts together to form a final test grade, what would the variance of the final test grades be?

(ii) What would the variance of Part A - Part B be?

$$\text{(i)}\quad Var\ (A + B) = 25 + 49 + 2*0.6*\sqrt{25}*\sqrt{49}$$
$$= 116$$

$$\text{(ii)}\quad Var\ (A - B) = 25 + 49 - 2*0.6*\sqrt{25}*\sqrt{49}$$
$$= 32$$

Ch 5. Probability

- Basic Concepts – Review of Set Theory and Venn Diagram
- Counting – Ordered, Unordered, Sampling With and Without Replacement.
- Probability Theory
- Base Rate: Bayes' Theorem

▪ Set Theory - Review

A set is a collection of elements.

Eg: A = {Head, Tail}, Head $\in$ A, Tail $\in$ A

B = {1, 2, 3, 4, 5, 6}

C = {1, 3, 5}

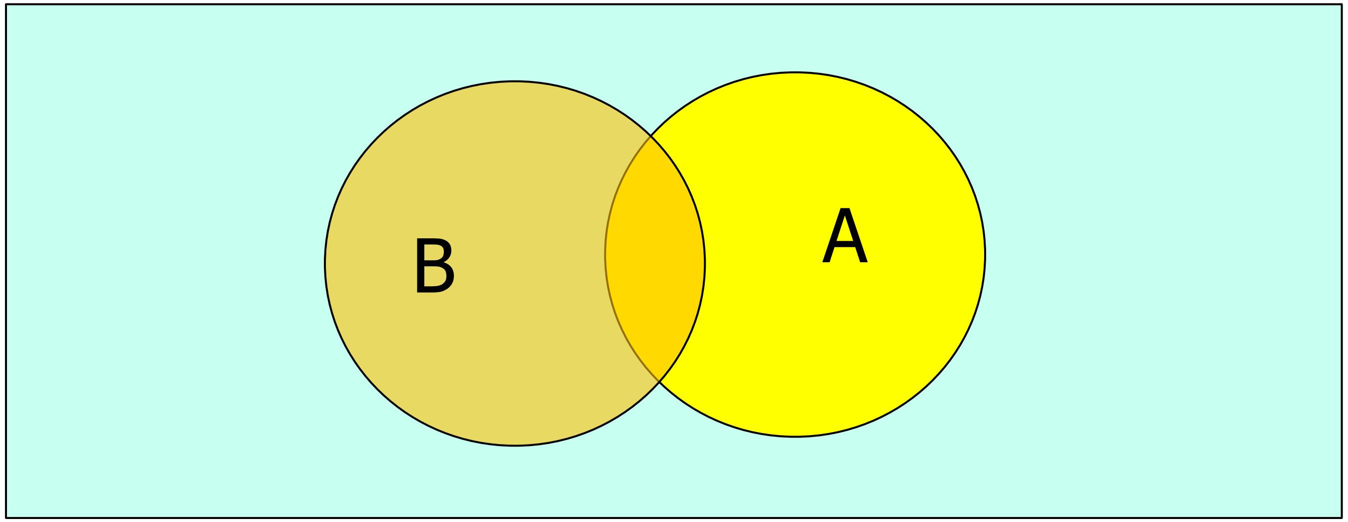
Set C is a subset of Set B since all elements in C are also in B. Notation: C $\subset$ B

The set of all elements in a given context is known as an universal set **Ω**.

Null set 0 is a set with no element.

- ## Venn Diagram

universal set $\mathbf{\Omega}$

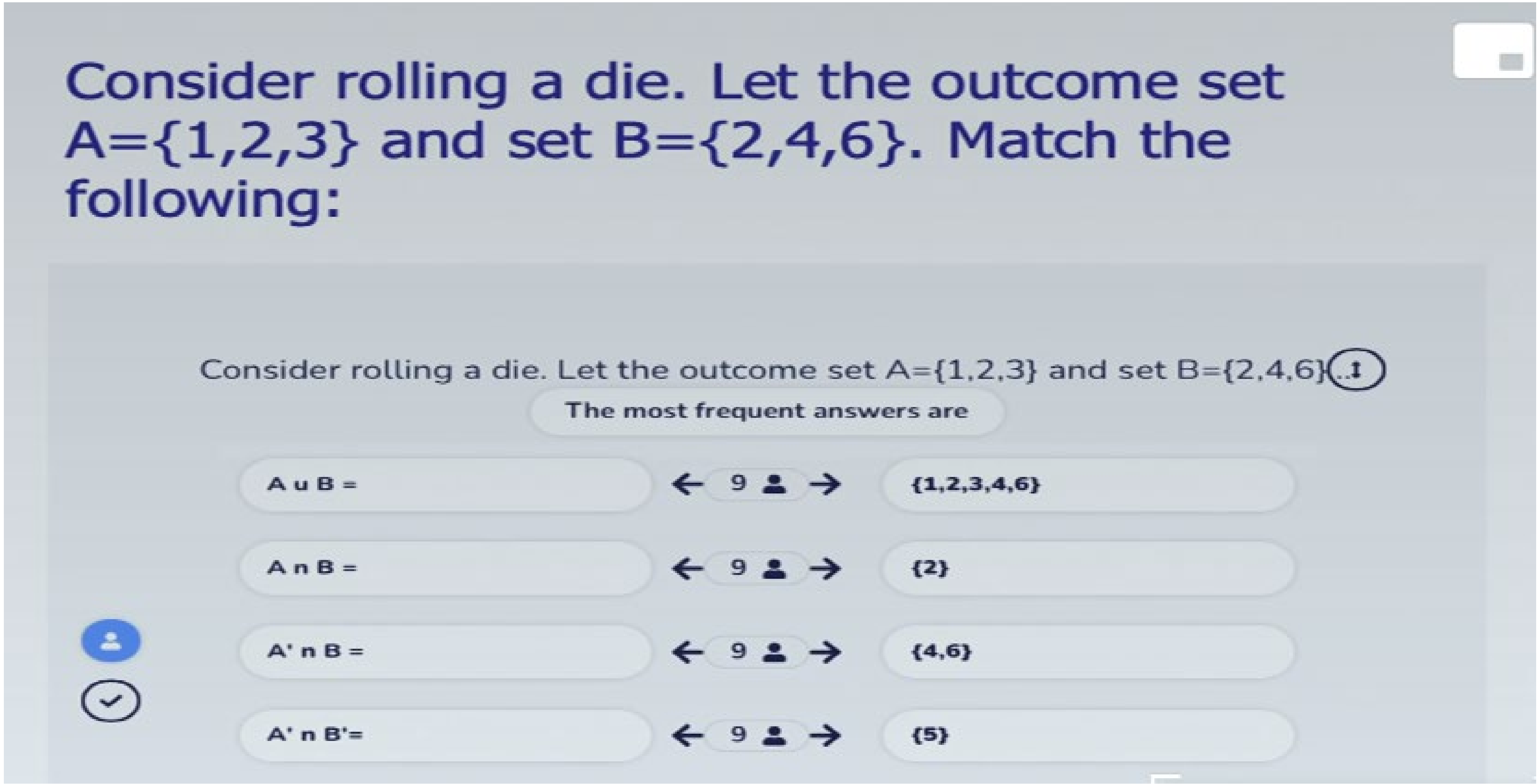


Set A $\subset$ $\mathbf{\Omega}$. Complement of A, denoted by A′, consists of all elements in $\mathbf{\Omega}$ that are not in A.
Eg: $\mathbf{\Omega}$ = {1,2,3,4,5,6}, A={1,2}, A′={3,4,5,6}

Union of A and B consists of all elements in A or B. Notation: A $\cup$ B.
Eg: {1,2,3} $\cup$ {2,3,4} = {1,2,3,4}

Intersection of A and B consists of all elements in both A and B. Notation: A $\cap$ B.
Eg: {1,2,3} $\cap$ {2,3,4} = {2,3}

Consider rolling a die. Let the outcome set A={1,2,3} and set B={2,4,6}. Match the following:



- Counting

When solving probability problems, we may involve counting.

Eg: Calculating the probability of guessing the correct 4 digits numeric code.

Each digit is sampled from $\{0,1,2,\ldots,9\}$.

Different scenarios for consideration:

- sampling with replacement
- sampling without replacement
- digits drawn are ordered
- digits drawn are unordered

- Counting – general scenario

Sampling	Eg: Pick 3 numbers (one at a time) from a set of 10 digits. Arrangement can be ordered or unordered	Two options: with or without replacement. After picking each number from the set, the number can be replaced or no replacement.

- Counting – ordered sampling with replacement

Ordered sampling k elements from a set of n elements with replacement.

No. of possible arrangements = n^k

Eg: Calculate the total number of different 4-digit numeric codes. Each digit is independently selected from {0,1,2,…,9}.

1st digit: 10 possible numbers

2nd digit: also 10 possible numbers.

3rd and 4th digit: each also 10 possible numbers.

Total arrangements = 10 x 10 x 10 x 10 = 10^4

- Counting – ordered sampling without replacement

Ordered sampling k elements from a set of n elements without replacement.

No. of possible arrangements =

n x $(n-1)$ x $(n-2)$ x $(n-3)$ x … x $(n-k+1)$

Number of k permutations of n elements:

$$^{n}\mathrm{P}_{k} = \frac{n!}{(n-k)!}$$

Eg: Calculate the total number of different 4-digit numeric codes. Each digit is sampled from {0,1,2,…,9} without replacement.

1^{st} digit: 10 possible numbers

2^{nd} digit: 9 possible numbers

3^{rd} digit: 8 possible numbers and so on.

Total arrangements = 10 x 9 x 8 x 7

i.e. there are ${}^{10}P_4 = \frac{n!}{(n-k)!} = \frac{10!}{(10-4)!}$ arrangements

Eg: 4 boys and 2 girls sit in a row. Find the following:

(i) No. of ways of putting these 6 people in a row.

(ii) No. of ways such that each girl has a boy to her left and to her right.

- Counting – unordered sampling without replacement

We choose k elements from a set of n elements and the ordering does not matter.

Number of arrangements equals to k combinations of n elements

$$= \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Note: $\binom{n}{k}$

$$= \frac{n!}{k!\,(n-k)!}$$

$$= \binom{n}{n-k}$$

Eg: How many combinations of two numbers between 1 and 6 are there:

Ans: $\frac{6!}{2!\,(6-2)!} = \frac{6 \times 5}{2 \times 1} = 15$

- Counting – unordered sampling with replacement

Pick k elements from a set of n elements, one at a time with replacement and the ordering does not matter.

Eg: Pick two numbers from {1, 2, 3}, we have

6 arrangements with n=3 and k=2.

i.e. (1,1), (1,2), (1,3), (2,2), (2,3) and (3,3)

In general:

No. of arrangements = $\binom{n+k-1}{k}$

Eg: A 6-bit code is made up of 4 1’s and 2 0’s.
Calculate:

(i) the no. of distinct codewords in the code?

(ii) the no. of codewords such that a ‘0’ has a ‘1’ to its left and to its right.

- Probability Theory

In a random experiment, the outcome of the experiment occurs with a certain probability.

Eg: We toss a coin. If the coin is unbiased, then the chances of getting a Head is 50%.

Eg: We roll a fair die two times. The outcome $\in \{11, 12, 13, 14, 15, 16, 21, 22, \ldots, 65, 66\}$.

Total of 36 possible outcomes.

Probability of getting a "1" followed by another "1" is 1/36.

Probability of getting 1^{st} no. = 2^{nd} no. is 6/36.

Definitions

- A sample space is the set S of all possible outcomes of an experiment.
- An event is a set of one or more (favorable) outcomes in the sample space.
- Two events are mutually exclusive if they have no outcomes in common.
- They are exhaustive if they cover all possible outcomes.
- Two events are independent if the probability that one occurs is not affected by whether or not the other has occurred.

Ch 5. Probability

- Basic Concepts – Review of Set Theory and Venn Diagram
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- Probability Theory
- Base Rate: Bayes' Theorem

Axioms of Probability:

(1) Probability of any outcome or event X is a non-negative:

$$P(X) \geq 0$$

(2) Probability of the sample space S is 1:

$$P(S) = 1$$

(3) If X_1, X_2, X_3, … are mutually exclusive events, then:

$$P(X_1 \text{ or } X_2 \text{ or } X_3 \ldots) = P(X_1) + P(X_2) + P(X_3) + \ldots$$

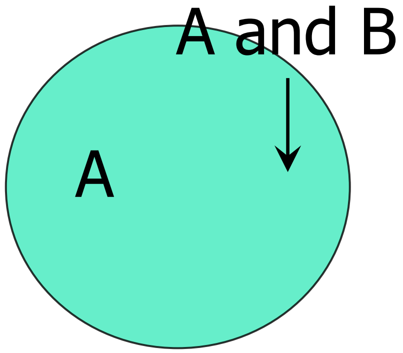
Probabilities and Events (mutually exclusive)

Eg: A company has decided that in the next 5 years, 40% of their new employees will be men, 30% will be Singaporean, and 35% will be foreigner women. What percentage of new employees will be Singaporean men?

	Men	Women	Total
Foreigner		0.35	
Singaporean	$0.3-0.25$ $=0.05$	$0.6-0.35$ $=0.25$	0.3
Total	0.4	$1-0.4$ $=0.6$	

Probabilities and Events (non-mutually exclusive)

Eg: A certain kind of fruit is grown in 2 districts, A and B. Both areas sometimes get fruitflies. Suppose the probabilities are P(A)=0.1, P(B)=0.05 and P(A and B)=0.02, what is the probability that one or other (or both) districts are infected at a given time?



P(A or B) = P(A) + P(B) – P(A and B)
= 0.1 + 0.05 – 0.02
= 0.13

Probabilities and Events (independent)

Eg: A company has 2 guards. Each carries a pager activated by sensors. Guard 1 and guard 2 respond to pager alert 80% and 50% of the time respectively. They independently report any alert. What is the probability that at least one will report an alert?

Worked Examples considered so far:

Probabilities and Events (mutually exclusive)

Probabilities and Events (non-mutually exclusive)

Probabilities and Events (independent)

Next: Consider Conditional Probabilities

i.e. Probability of an event A given that an event B has occurred, denoted as $P(A|B)$

Conditional Probabilities

Assuming that 10 percent of the days are rainy in a certain city.

$$P(\text{rain}) = 0.1$$

The probability that it rains given that it is cloudy might be, say,

$$P(\text{rain}|\text{cloudy}) = 0.8$$

This is known as conditional probability: the probability of event A given that event B has occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{where } B > 0$$

Conditional Probabilities

Given P(A)=0.5 and P(B)=0.8, determine the following.

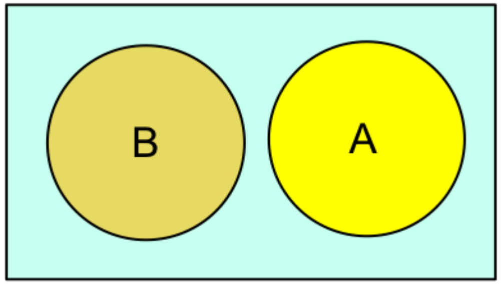


Fig 1.

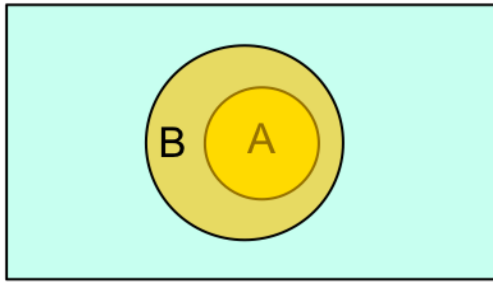


Fig 2.

(1) P(A|B) for Fig 1.

(2) P(A|B) for Fig 2.

(3) P(B|A) for Fig 2.

Conditional Probabilities

Eg: A jar contains 10 blue, 5 red, 4 green and 1 yellow marbles. Two marbles are randomly picked. What is the probability that one will be blue and the other yellow?

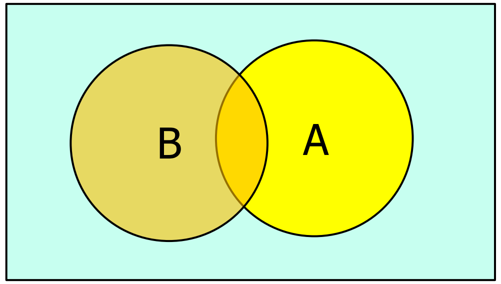
Illustration using a tree diagram:

Bayes' Theorem

Given two events A and B, where P(A) > 0, we have:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

Note: $P(A) = P(A \cap B) + P(A \cap B')$



Similarly:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Bayes' Theorem

Eg: A test correctly identifies a disease in 95% of people who have it. It correctly identifies no disease in 94% of people who do not have it. In the population, 3% of the people have the disease. What is the probability that one has the disease if tested positive?

Summary For Probability Calculations

Are X and Y mutually exclusive?

- **Yes**
  1. P(X and Y) = 0 because event (X and Y) is impossible
  2. P(X or Y) = P(X) + P(Y)
- **No**
  1. P(X and Y) > 0
  2. P(X or Y) = P(X) + P(Y) – P(X and Y)

  Are X and Y independent?

  - **Yes**
    1. P(X|Y) = P(X)
    2. P(X and Y) = P(X) P(Y)
  - **No**
    1. $P(X|Y) = \frac{P(X \text{ and } Y)}{P(Y)}$
    2. P(X and Y) = P(X|Y) P(Y)
       = P(Y|X) P(X)